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IoT Based Smart Environmental Monitoring System

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

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BACHELOR OF ENGINEERING

IN

CSE, AI &DS

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DECLARATION

We, **R. Indhu, K. Niharika** of the **B.E CSE, AI&DS** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled “**IoT Based Smart Environmental Monitoring System**” is the result of my own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: SIMATS ENGINEERING, THANDALAM

Date: 27.06.2026

Signature of the students with names



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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**IoT Based Smart Environmental Monitoring System**” has been carried out by **R. Indhu, K. Niharika** under the supervision of **Dr. VIJAYARAGAVAN, Dr. S. K. ARUNA** and is submitted in partial fulfilment of the requirements for the current semester of the **B.E CSE, AI&DS** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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Signature With Student Name

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ABSTRACT

The IoT Based Smart Environmental Monitoring System is designed to monitor environmental parameters such as temperature, humidity, soil moisture, rainfall, and air quality in real time using IoT technology. The system utilizes various sensors connected to a Node MCU microcontroller to collect environmental data and transmit it to a cloud platform through Wi-Fi communication. Users can remotely access the collected information through a web dashboard or mobile application, enabling continuous monitoring from any location.

The system provides timely alerts when environmental conditions exceed predefined thresholds, allowing quick response to potential hazards. By automating data collection and monitoring, the solution reduces manual effort, improves accuracy, and enhances decision-making. The integration of IoT technology ensures efficient communication between sensors, processing units, and cloud services.

The proposed system is cost-effective, user-friendly, and suitable for applications such as smart agriculture, weather monitoring, pollution control, industrial safety, and smart city development. Overall, the project demonstrates how IoT and sensor technologies can be combined to create an intelligent environmental monitoring solution that promotes sustainability, safety, and efficient resource management.

CHAPTER 1

INTRODUCTION

1.1 Background Information

The IoT Based Smart Environmental Monitoring System is developed to monitor environmental conditions such as temperature, humidity, soil moisture, rainfall, and air quality using sensors and IoT technology. Traditional environmental monitoring methods often require manual observation and data recording, which can be time-consuming and less accurate. This system automates the monitoring process and provides real-time environmental data.

The collected data is transmitted to a cloud platform through a Node MCU microcontroller with Wi-Fi connectivity. Users can access the information remotely using a mobile application or web dashboard. The system also generates alerts when environmental conditions exceed predefined thresholds.

The system is designed to be low-cost, efficient, and user-friendly. It is suitable for applications such as smart agriculture, weather monitoring, pollution control, and smart city management.

1.2 Project Objectives

The main objective of this project is to develop an IoT-based system capable of monitoring environmental conditions in real time. The system aims to collect data from sensors and transmit it to cloud platforms for remote monitoring.

Another objective is to provide alert notifications when environmental parameters exceed safe limits. The project also focuses on improving environmental management through automation and accurate data collection.

Overall, the system is designed to be reliable, cost-effective, and useful for agriculture, industries, and environmental protection.

1.3 Significance

The significance of this project lies in its ability to provide continuous environmental monitoring without human intervention. It improves accuracy, reduces manual effort, and enables quick responses to environmental changes.

The system helps farmers monitor crop conditions, industries track air quality, and authorities manage environmental resources effectively. It also supports sustainable development through efficient environmental monitoring.

1.4 Scope

The scope of this project includes monitoring temperature, humidity, soil moisture, rainfall, and gas levels using sensors connected to a NodeMCU microcontroller. The collected data is displayed on a cloud platform for remote access.

The system is suitable for agricultural fields, industries, weather stations, and smart city applications. Future enhancements may include AI-based prediction systems and integration with smart automation devices.

1.5 Methodology Overview

The methodology begins with collecting environmental data using multiple sensors. The Node MCU microcontroller processes the sensor readings and sends them to a cloud platform through Wi-Fi communication.

The data is stored and displayed on a dashboard where users can monitor environmental conditions. Alerts are generated when abnormal conditions are detected. Finally, the system is tested to ensure accurate data collection and reliable performance.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

Environmental conditions continuously change and can significantly affect agriculture, industries, and public health. Traditional monitoring methods rely on manual observations, which are often inefficient and prone to errors.

Lack of real-time monitoring can result in delayed responses to environmental hazards such as pollution, drought, or harmful gas leakage. Therefore, an automated monitoring system is required.

2.2 Evidence of the Problem

Farmers often face crop losses due to insufficient monitoring of soil moisture and weather conditions. Industries may experience safety issues because of delayed detection of harmful gases.

Research studies show that real-time environmental monitoring improves resource management and reduces environmental risks.

2.3 Stakeholders

- Farmers
- Environmental agencies
- Industrial organizations
- Researchers
- Smart city authorities
- General public

2.4 Supporting Data/Research

Studies in IoT and environmental science demonstrate that sensor-based monitoring systems provide accurate and continuous environmental data. Cloud-based monitoring platforms enable remote access and effective decision-making. Research also highlights the importance of real-time alerts in preventing environmental damage and improving public safety.

CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process

The project was developed using a Node MCU microcontroller and environmental sensors. Each sensor was connected and calibrated to collect accurate environmental data.

The cloud platform was configured to receive and display sensor readings. Testing was conducted to ensure proper communication between hardware and cloud services.

3.2 Tools and Technologies Used

Software

- Arduino IDE
- Thing Speak / Blynk
- Embedded C

Hardware

- Node MCU ESP8266
- DHT11 Sensor
- Soil Moisture Sensor
- Rain Sensor
- MQ-2 Gas Sensor
- Jumper Wires
- Breadboard
- Power Supply

3.3 Solution Overview

The proposed system collects environmental data through sensors connected to the Node MCU controller. The controller processes the sensor readings and uploads the data to a cloud platform using Wi-Fi.

Module 1: Data Acquisition

Key Features:

- Temperature monitoring
- Humidity monitoring
- Soil moisture detection
- Rain detection
- Gas leakage monitoring

Module 2: Cloud Monitoring and Alerts

Key Features:

- Real-time data upload
- Cloud dashboard visualization
- Remote monitoring
- Threshold-based alerts
- Data logging and analysis

3.4 Engineering Standards Applied

The project follows Software Development Life Cycle (SDLC) principles for design and implementation. Sensor interfacing standards and IoT communication protocols are used to ensure reliability and accuracy.

Proper coding practices, testing procedures, and hardware integration methods are followed to achieve efficient system performance.

3.5 Solution Justification

The proposed solution provides continuous monitoring, reduces manual effort, and enables remote access to environmental data. It is affordable, scalable, and easy to implement.

The integration of IoT technology allows users to receive real-time updates and alerts, making the system effective for environmental management applications.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results

The system successfully monitored environmental parameters and uploaded data to the cloud platform. Sensor readings were displayed accurately on the dashboard.

Real-time monitoring and alert features operated effectively under testing conditions.

4.2 Challenges Encountered

- Sensor calibration issues
- Network connectivity interruptions
- Environmental noise affecting readings
- Power management challenges

4.3 Possible Improvements

Future improvements may include:

- AI-based environmental prediction
- Solar-powered operation
- Mobile application integration
- Advanced analytics dashboard

4.4 Recommendations

The system should be deployed in larger environmental monitoring applications. Additional sensors can be integrated for more comprehensive analysis and improved accuracy.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Key Learning Outcomes

Academic Knowledge

- IoT fundamentals
- Sensor interfacing
- Environmental monitoring concepts
- Cloud computing basics

Technical Skills

- Node MCU programming
- Arduino IDE development
- Sensor calibration
- Data communication

5.2 Challenges Encountered

Developing the project required solving hardware integration issues and ensuring reliable communication between sensors and cloud services. Network connectivity and sensor accuracy were major challenges.

5.3 Insights into the Industry

The project provided knowledge about real-world IoT applications used in agriculture, environmental protection, and smart city infrastructure. It demonstrated the importance of automation and data-driven decision-making.

5.4 Conclusion of Personal Development

This project improved technical skills, problem-solving abilities, teamwork, and project management capabilities. It provided valuable practical experience in IoT system development and environmental monitoring technologies.

Chapter 6

Implementation

APPENDIX A – PSEUDOCODE

START

Initialize Node MCU

Initialize Sensors

Connect Wi-Fi

WHILE system is running

Read Temperature

Read Humidity

Read Soil Moisture

Read Rain Sensor

Read Gas Sensor

Upload Data to Cloud

IF value exceeds threshold

Send Alert

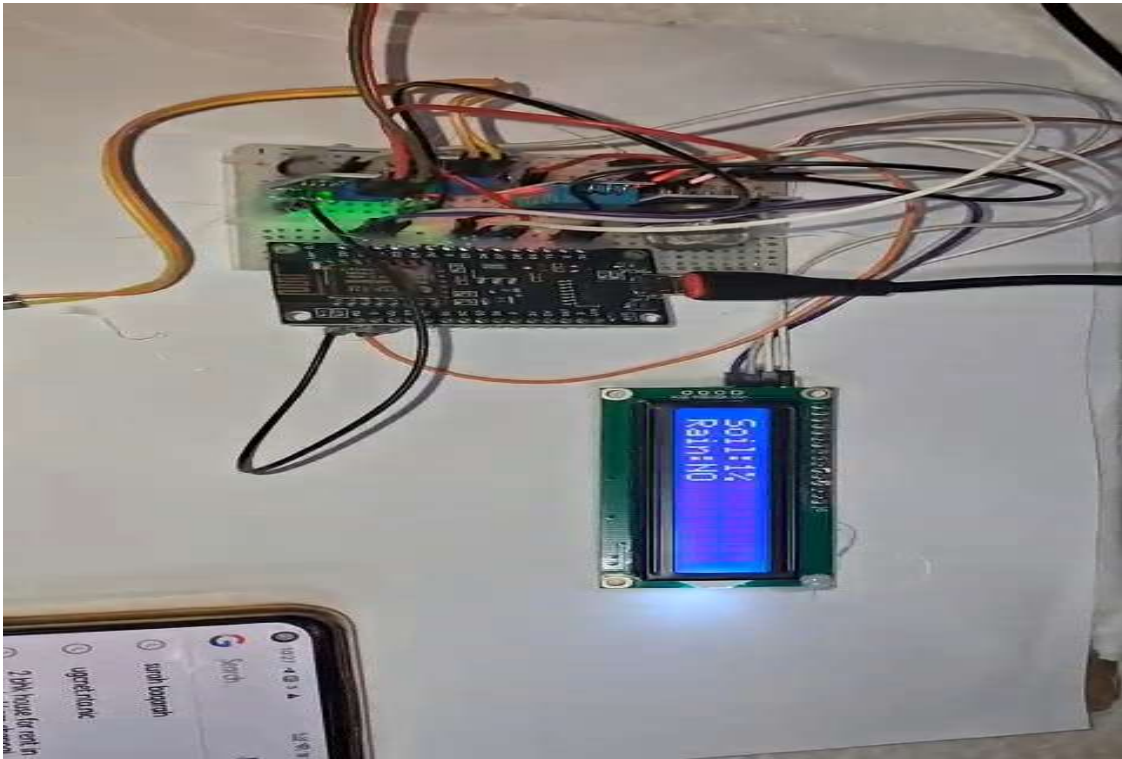
END IF

Display Sensor Values

END WHILE

STOP

Working Model:



Python Code:

```
from machine import Pin, ADC

from dht import DHT11

import time

Sensors

soil = ADC(0)

rain = Pin(14, Pin.IN)    # D5

mq2 = Pin(12, Pin.IN)    # D6

dht = DHT11(Pin(13))    # D7

while True:
```

```
# Soil Moisture

soil_raw = soil.read()

soil_percent = int((1023 - soil_raw) * 100 / 1023)

# Rain Sensor

rain_detected = rain.value() == 0

# MQ2 Gas Sensor

gas_detected = mq2.value() == 0

# DHT11

dht.measure()

temperature = dht.temperature()

humidity = dht.humidity()

print("----- Environmental Data -----")

print("Soil Moisture:", soil_percent, "%")

print("Rain:", "YES" if rain_detected else "NO")

print("Temperature:", temperature, "C")

print("Humidity:", humidity, "%")

print("Gas:", "DETECTED" if gas_detected else "SAFE")

# Alerts

if soil_percent < 30:

    print("ALERT: Soil Dry")

if rain_detected:

    print("ALERT: Rain Detected")
```

```
if temperature > 30:
    print("ALERT: High Temperature")
if humidity < 40:
    print("ALERT: Low Humidity")
if gas_detected:
    print("ALERT: Gas Detected")
print("-----")
time.sleep(2)
```

Chapter 7

CONCLUSION

The IoT Based Smart Environmental Monitoring System successfully demonstrates the use of IoT technology for real-time environmental monitoring. By integrating sensors, Node MCU, and cloud platforms, the system provides accurate and continuous monitoring of environmental parameters.

The project improves efficiency, reduces manual effort, and enables quick responses to environmental changes through remote monitoring and alert mechanisms. Overall, the system offers a reliable, cost-effective, and scalable solution for environmental management, agriculture, industrial safety, and smart city applications.

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